

COMP90077

Lecture 8

LP Rounding

William Umboh
University of Melbourne

Caution

- Slides are designed as presentation aid to go along with lecture, not as study material
- Slides not designed to make sense on their own

Today

- Last time: Deterministic LP Rounding for Vertex Cover
- Deterministic LP Rounding for Set Cover
- Randomized LP Rounding for Set Cover and Min (s,t)-Cut

Recap: LP

Linear Program

Linear Program

n variables, m constraints

Coefficients: $[a_1, \dots, a_n] = a$

$$\begin{array}{|c|} \hline b_{1,1}, b_{1,2}, b_{1,3}, \dots, b_{1,n} \\ \hline \vdots \\ \hline b_{m,1}, b_{m,2}, \dots, b_{m,n} \\ \hline \end{array} \quad \begin{array}{|c|} \hline c_1 \\ \hline \vdots \\ \hline c_m \\ \hline \end{array}$$

$B \qquad c$

Variable vector x

Integer Linear Program

Linear Program

$$\max \sum_i a_i \cdot x_i$$

$$\text{s.t. } b_{1,1}x_1 + \dots + b_{1,n}x_n \leq c_1$$

$\vdots$

$$b_{m,1}x_1 + \dots + b_{m,n}x_n \leq c_m$$

$$\max \quad a^T x$$

$$\text{s.t. } Bx \leq c$$

$$x \in \{0,1\}^n$$

$$x \in [0,1]^n$$

Recall: LP Facts

- Integer programs are NP-hard
- Linear programs can be solved in time polynomial in:
 - n = number of variables
 - L = number of bits to represent coefficients (a , B , c)
 - m = number of constraints
- Given an integer program and its linear program
 - For minimization, $\text{OPT-LP} \leq \text{OPT-IP}$
 - For maximization, $\text{OPT-LP} \geq \text{OPT-IP}$

LP Rounding Framework (Minimization)

To get α -approximation

1. Write equivalent integer program
 - optimal solution for original problem = optimal solution for IP
2. Solve linear program
3. Round optimal LP solution x^* into an integral solution X
4. Analysis:

$$\text{Cost of } X = \sum_v w_v X_v \leq \alpha \sum_v w_v x_v^* = \alpha \text{ OPT-LP} \leq \alpha \text{ OPT-IP}$$

Recap: Vertex Cover LP Rounding

- 2-approximation for Vertex Cover
- Easily generalizes to Weighted Vertex Cover
- Key property of Vertex Cover LP Rounding Algorithm

Linear Program for Vertex Cover (Weighted)

$$\min \sum_v x_v \cdot w_v$$

$$\text{s.t. } x_u + x_v \geq 1 \quad \forall (u,v) \in E$$

$$x_v \geq 0 \quad \forall v$$

For each vertex, variable x_v

$$x_v = \begin{cases} 1 & \text{if } v \in C \\ 0 & \text{if } v \notin C \end{cases}$$

x^* is optimal soln to LP

$$\sum_v x_v^* \leq |C^*|$$

where C^* is min vertex cover.

$$\forall (u,v) \quad x_u + x_v \geq 1$$

$$\Rightarrow x_u \geq \frac{1}{2} \text{ or } x_v \geq \frac{1}{2}$$

$$C = \{v : x_v \geq \frac{1}{2}\}.$$

Key Property

$$\text{Let } X_v = \begin{cases} 1 & \text{if } v \in C \\ 0 & \text{o/w} \end{cases}$$

$\Rightarrow \forall \text{ weights } w,$

$$\sum_v w_v X_v \leq 2 \sum_v w_v x_v^*$$

For every v , $X_v \leq 2x_v^*$

Weighted Set Cover – Deterministic Rounding

Set Cover (Weights)

n elements U

m sets $S_1, \dots, S_m \subseteq U$

weights $w_1, \dots, w_m \geq 0$

$$\min \sum_i x_i w_i$$

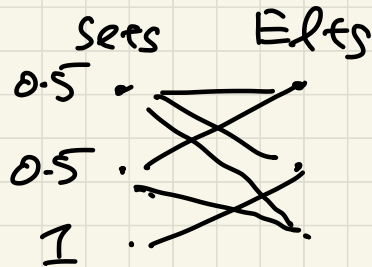
$$\text{s.t. } \sum_{S_i \ni e} x_i \geq 1 \quad \forall e \in U$$

$$x_i \geq 0 \quad \forall i$$

Def (Frequency)

$$\text{Frequency } f = \max_{e \in U} |\{S_i \ni e\}|$$

Optimal fractional solution x^*



$$I = \{i : x_i^* \geq \frac{1}{f}\}$$

Let X be indicator for I

$$\forall i \quad x_i \leq f \cdot x_i^*$$

$$\Rightarrow \sum_{i \in I} w_i = \sum_i w_i x_i \leq f \cdot \sum_i w_i x_i^*$$

Weighted Set Cover – Randomized Rounding

Probability preliminaries

- Linearity of Expectation: For any random variables A, B , we have

$$E[A + B] = E[A] + E[B]$$

- Independence: If X, Y are independent events, then

$$\Pr[X \wedge Y] = \Pr[X] \cdot \Pr[Y]$$

Union bound. $\Pr[X \vee Y] \leq \Pr[X] + \Pr[Y]$

Randomized $O(\log n)$ -approx for Set Cover
Goal Produce random I s.t.

$$E\left[\sum_{i \in I} w_i\right] \leq O(\log n) \sum_i w_i x_i^*$$

$x_i^* \in [0, 1] \Rightarrow$ Interpret as probabilities.

Alg 1

For every i , add i to I v.p. x_i^* indep.

Let X be indicator for I

$$E\left[\sum_{i \in I} w_i\right] = E\left[\sum_i X_i \cdot w_i\right]$$

$$= \sum_i E[X_i] w_i$$

$$= \sum_i \Pr[X_i = 1] w_i = \sum_i x_i^* w_i$$

Feasibility Want to ensure every element covered.

Fix $e \in U$

$$\Pr[e \text{ not covered}] = \Pr\left[\bigwedge_{S_i \ni e} S_i \text{ not chosen}\right]$$

$$= \prod_{S_i \ni e} \Pr[S_i \text{ not chosen}]$$

$$= \prod_{S_i \ni e} (1 - \alpha x_i^*)$$

$$\leq \prod_{S_i \ni e} \exp(-\alpha x_i^*)$$

$$= \exp\left(-\underbrace{\sum_{S_i \ni e} (\alpha x_i^*)}_{\geq 1}\right) \leq \exp(-\alpha)$$

$$\Pr[\exists e \text{ not covered}] \leq \sum_{e \in U} \Pr[e \text{ not covered}] \leq n \cdot \exp(-\alpha)$$

Probability Amplification

Alg 2

For every i , add i to $\hat{I}$ w.p. x_i^* indep.

Want to ensure $\Pr[\text{every element covered}] \geq \Omega(1)$

Want α s.t. $n \cdot \exp(-\alpha) \leq O(1)$

$$n \cdot e^{-\alpha} \leq O(1)$$

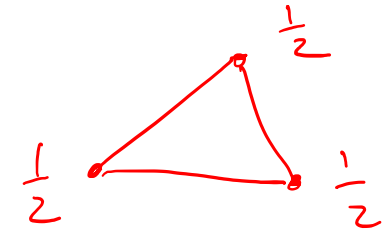
$$\alpha = \ln n + 1 \Rightarrow n \cdot e^{-\alpha} \leq n \cdot e^{-\ln n - 1} \\ = e^{-1}$$

$$\alpha = c \cdot \ln n \Rightarrow n \cdot e^{-c \ln n} = n \cdot n^{-c}$$

Alteration $\hat{I}$, for each $e \in U$ not covered by $\hat{I}$
add cheapest set containing e .

Integrality Gap

- Are these the best possible rounding schemes?
- Integrality gap of LP formulation = $\max \text{OPT-IP} / \text{OPT-LP}$ over all instances
- The smaller the gap, the better the LP formulation
- Tutorial 6: Integrality gap of Vertex Cover LP $\geq 4/3$
- Can we make it worse?
- Thm. Integrality Gap of Unweighted Set Cover LP $\geq \Omega(\log n)$



$$\text{OPT-LP} = \frac{3}{2}$$

$$\text{OPT-IP} = 2$$

$$\text{Gap} \geq \frac{4}{3}$$

n vertices

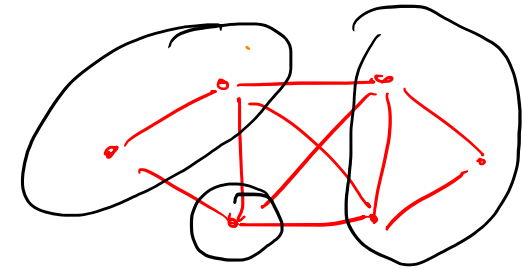
$$x_v = \frac{1}{2}$$

Complete graph

$$\text{Vertex Cover} \geq n-1$$

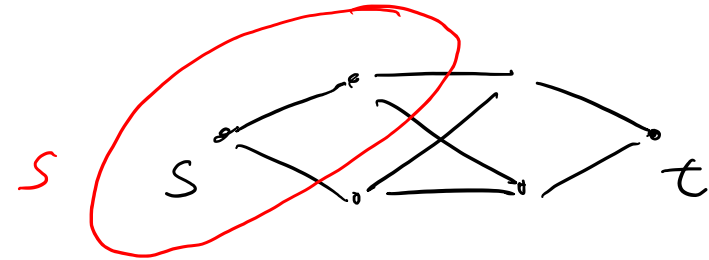
$$\frac{n-1}{n/2} = 2(1 - \frac{1}{n})$$

Graph Partitioning Problems



- General template:
 - Given a graph G with edge costs, partition G into pieces to minimize cost of cut edges
- Applications:
 - Clustering (each cut is a cluster, edge cost represents likelihood the endpoints are in the same cluster)
 - Divide and conquer algorithms
 - Dual to max flow problems
 - Connections to metric embeddings, graph theory, etc

Min (s,t)-cut



- The most basic graph partitioning problem
- Dual to max s-t flow
- Given $G = (V, E)$ with edge costs c_e and two vertices s, t ,
 - partition V into two pieces S, T such that $s \in S$ and $t \in T$ and want to minimize cost of cut edges
 - Denote cost as $c(S)$

Min (s,t)-cut

$G = (V, E)$, edge costs c_e , vertices s, t .

Want $S \subseteq V$ s-t. $s \in S$, $t \notin S$

minimize $\sum_{e \in \delta(S)} c_e$

$\delta(S) = \{(u,v) \in E : |\{u,v\} \cap S| = 1\}$.

LP1 min $\sum_e x_e \cdot c_e$

$x_e = \begin{cases} 1 & \text{if } e \in \delta(S) \\ 0 & \text{o/w} \end{cases}$

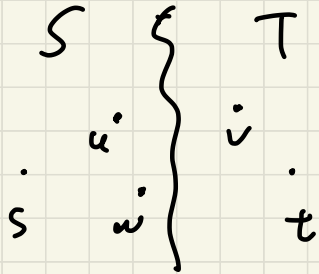
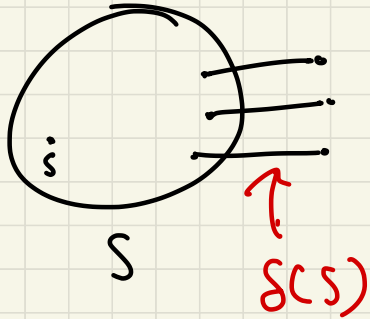
s-t. $\sum_{e \in P} x_e \geq 1 \quad \forall P \in \mathcal{P}_{s,t}$

$\mathcal{P}_{s,t} = \{ \text{s-t paths} \}$

LP2 min $\sum_{u,v} d_{uv} \cdot c_{uv}$

$d_{uv} = \begin{cases} 1 & \text{if cut} \\ 0 & \text{o/w} \end{cases}$

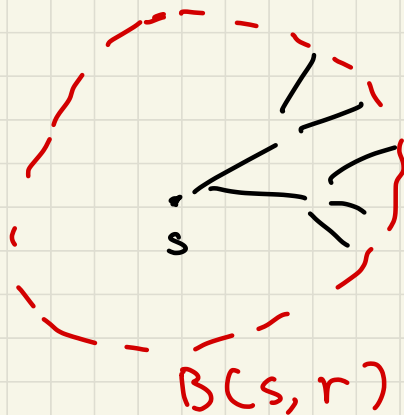
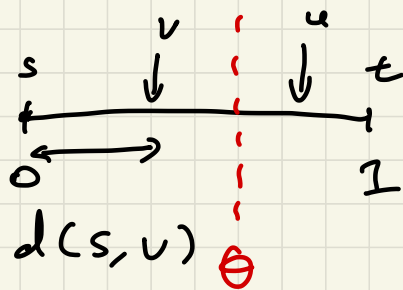
s-t. $d_{uv} \leq d_{uw} + d_{wv} \quad \forall u,v,w$
 $d_{st} = 1$



$$\underline{LP2} \quad \min \sum_{u,v} d_{u,v} \cdot c_{u,v}$$

$$\text{s.t. } d_{uv} \leq d_{uw} + d_{wv} \quad \forall u, v, w$$

$$d_{st} = 1$$

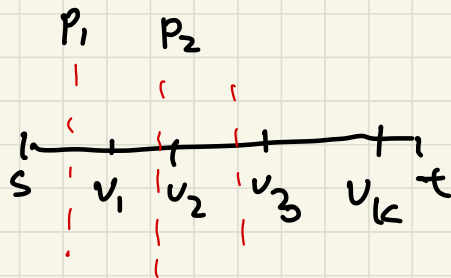


Pick $\Theta \in [0, 1)$ u.ar.

$$S = \{ v : d(s, v) \leq \Theta \}.$$

$$\Pr[u, v \text{ cut}] = d(s, u) - d(s, v) \leq d(u, v)$$

$$E[c(S)] = \sum_{u,v} \Pr[u, v \text{ cut}] \cdot c_{u,v} \leq \sum_{u,v} d(u, v) \cdot c_{u,v}$$



$$S_i = \{s, v_1, \dots, v_i\}.$$

$$E[c(S)] = \sum_i p_i \cdot c(S_i) \leq \text{OPT-LP}$$

$$\Rightarrow \exists i \text{ s.t. } c(S_i) \leq \text{OPT-LP}$$

$\Rightarrow$ Deterministic exact algorithm
for min (s, t) -cut

$$\text{Integrality gap} = 1$$

Summary

To get α -approximation

1. Write equivalent integer program
 - optimal solution for original problem = optimal solution for IP
2. Solve linear program
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4. Analysis:

$$\text{Cost of } X = \sum_v w_v X_v \leq \alpha \sum_v w_v x_v^* = \alpha \text{ OPT-LP} \leq \alpha \text{ OPT-IP}$$

Summary

- Key Property for Rounding:
 - For every variable j , $X_j \leq \alpha x_j^*$ (deterministic)
 - For every variable j , $E[X_j] \leq \alpha x_j^*$ (randomized)
- Applications
 - Vertex Cover
 - Set Cover
 - Min (s,t) cut